

Basic Python Interview Questions

General & Basics

1. What is Python? What are its key features?

- *Answer:* Interpreted, dynamically typed, object-oriented, high-level, easy syntax, extensive standard library.

2. Is Python an interpreted or compiled language?

- *Answer:* Python is both. Source code is compiled into byte code, which is then interpreted by the Python Virtual Machine (PVM).

3. What is PEP 8?

- *Answer:* The style guide for Python code.

4. How is memory managed in Python?

- *Answer:* Private heap space, Python memory manager, and garbage collector - primarily reference counting.

5. What are the differences between Python 2 and Python 3?

- *Answer:* Print as a function, integer division, Unicode strings by default, etc.

Data Types & Structures

6. What is the difference between a list and a tuple?

- *Answer:* Lists are mutable, tuples are immutable.

7. What is a dictionary in Python?

- *Answer:* An unordered, mutable collection of key-value pairs.

8. What is a set? How is it different from a list?

- *Answer:* An unordered collection of unique elements. Sets do not allow duplicates and don't support indexing.

9. Explain the difference between mutable and immutable data types.

- *Answer:* Mutable objects can be changed after creation (lists, dicts, sets). Immutable objects cannot (strings, ints, tuples).

10. What is negative indexing?

- *Answer:* Using negative numbers to access elements from the end of a sequence, e.g., `list[-1]` gets the last element.

11. How do you copy an object in Python? (Shallow copy vs. Deep copy)

- *Answer:* Using `copy` module. Shallow copy creates a new object but inserts references into it. Deep copy recursively copies objects.

Control Flow & Functions

12. What is the difference between `break`, `continue`, and `pass`?

- *Answer:* `break` exits a loop, `continue` skips the rest of the current iteration, `pass` is a null operation/placeholder.

13. What are `*args` and `**kwargs`?

- *Answer:* `*args` allows passing a variable number of positional arguments. `**kwargs` allows passing a variable number of keyword/named arguments.

14. What is a lambda function?

- *Answer:* An anonymous, single-expression function created using the `lambda` keyword.

15. What is a generator in Python?

- *Answer:* A function that returns an iterator using the `yield` keyword instead of `return`, evaluating values lazily.

16. What is a decorator?

- *Answer:* A function that takes another function and extends its behavior without explicitly modifying it.

Object-Oriented Programming (OOP)

17. What is `__init__` ?

- *Answer:* The constructor method in Python classes, called when an object is instantiated.

18. What is the `self` keyword?

- *Answer:* It represents the instance of the class and is used to access variables and methods associated with that instance.

19. What are magic methods (dunder methods)?

- *Answer:* Methods with double underscores like `__str__`, `__len__`, `__add__` that allow overriding default Python behaviors.

20. Does Python support multiple inheritance?

- *Answer:* Yes. A class can inherit from multiple parent classes.

Built-in Functions & Modules

21. What is the purpose of `is` vs `==` ?

- *Answer:* `==` checks for value equality. `is` checks for object identity/memory location.

22. What does the `map()`, `filter()`, and `reduce()` functions do?

- *Answer:* `map` applies a function to all items, `filter` filters items based on a function, `reduce` applies a rolling computation to sequential pairs of values in a list.

23. What is a virtual environment in Python?

- *Answer:* An isolated environment for Python projects to manage dependencies separately.

24. How do you handle exceptions in Python?

- *Answer:* Using `try`, `except`, `else`, and `finally` blocks.

25. What does `if __name__ == "__main__": do?`

- *Answer:* It ensures that the block of code inside it runs only if the script is executed directly, not if it's imported as a module.

Common Basic Coding Questions

- Reverse a String
- Check if a string is a Palindrome
- Find the Factorial of a number (Iterative and Recursive)
- Check if a number is Prime
- Generate the Fibonacci sequence
- Swap two numbers without a temporary variable
- Implement Binary Search